

BATMAN HAT v1 for Raspberry Pi 4 – schematic generated from hardware/hat-v1/gen/design.py
Design spec: docs/design/hat-v1-spec.md (sections referenced in the part notes)

Blocks (one sheet each):

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1 power_in battery pads+SMBJ20CA TVS, TPS26631 eFuse + reverse FET, INA226 #1 (0x40), BAT_PRESENT
2 power_5v LM33640 5.17 V / 4 A with hardware UVLO on EN, LM74700 ideal diode to the Pi 5 V pins
3 power_3v3 LM74700 x2 OR-ing (battery / Pi 5 V), TPS29333F FCCM 3.33 V, INA226 #2 (0x41), pi filter
4 softpower LTC2955-2 push-button controller, AUTO-ON jumper, GPIO27 -> KILL, GPIO26 <- INT
5 pi_header 40-pin header, CAT24C32 IAT EEPROM (write-protected), mounting holes
6 halow mPCIE 5.2H socket for Wio-M56108, bulk capacitors
7 security SLB9672 TPM 2.0 on SPI1, AT97C608C-TFLXTL5 (0x36), RV-3028-C7 RTC (0x52) + supercap
8 gnss u-blox MAX-M105 on UART5 + PPS, active antenna bias, U.FL
9 debug Tag-Connect TC2050 (UART0 + HaLow SPI), test points with expected values, power flags

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Enable logic: 5 V buck runs when (LTC2955 on) AND (eFuse PGOOD) AND (VSYS > 6.40 V).
3.3 V buck runs when (LTC2955 on) OR (LTC2955 unpowered: USB-C-only bench) OR (BENCH jumper).
Tier A parts (TPM, ATECC608C, EEPROM, RTC, GNSS) are not assembled by JLCPCB: fitted in Taiwan (spec 5.9).
No test points on TPM / SPI1 nets (spec 5.1).

power_in

1 Power input: TVS, eFuse, reverse protection, system current

File: power_in.kicad_sch

softpower

4 Soft power: LTC2955-2 push-button / AUTO-ON / KILL

File: softpower.kicad_sch

security

7 TPM 2.0 SLB9672, ATECC608C, RTC RV-3028

File: security.kicad_sch

power_5v

2.5 V buck for the Pi + ideal-diode (HAT back-power rule)

File: power_5v.kicad_sch

pi_header

5 Pi 40-pin header, HAT ID EEPROM, mounting

File: pi_header.kicad_sch

gnss

8 GNSS MAX-M10S + active antenna bias

File: gnss.kicad_sch

power_3v3

3 3.3 V for HaLow: source OR-ing, FCCM buck, HaLow current

File: power_3v3.kicad_sch

halow

6 mPCIe socket for Wio-WM6108 (pinout unchanged)

File: halow.kicad_sch

debug

9 Debug: Tag-Connect, test points, LEDs, power flags

File: debug.kicad_sch

Label-based schematic: same net name = connected
Root: block overview

Batman project – generated by hardware/hat-v1/gen/gen_sch.py, do not edit by hand

Sheet: /
File: batman-hat.kicad_sch

Title: Batman HAT v1 (Pi 4) – replaces Seeed WM1302 HAT

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